A Decomposition-Aware Multi-Scale Mamba Model for Nonstationary Financial Forecasting

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Internal draft. The theory and data analysis are complete; final out-of-sample model results are pending.

Abstract

Financial level series combine persistent local levels with changes in location, scale, volatility, and cross-series dependence. We introduce PMS-Mamba, a decomposition-aware multi-scale patch Mamba model for this setting. The model first reversibly normalizes each input window, decomposes it into moving-average trend, smoothed-residual, and residual components, constructs several mean-pooled temporal resolutions, and applies bidirectional Mamba blocks to ordered temporal patches at each resolution. This design differs from inverted variable-token Mamba encoders: for a univariate task, their state-space scan receives a single variable token and cannot traverse temporal tokens. We establish that the implemented decomposition is additive, that its pooling stage is non-expansive in the maximum norm, and that fixed mean fusion remains coordinatewise bounded by the scale-specific forecasts. Selection-safe analysis of four daily financial panels—foreign exchange, U.S. Treasury yields, cross-asset prices, and cryptoassets—finds level nonstationarity under the prespecified ADF/KPSS diagnostic pair, validation mean shifts from 0.51 to 7.75 training standard deviations in magnitude, and first-difference volatility ratios from 0.82 to 4.52. These observations motivate, but do not prove, the model's inductive biases. The final 2023–2026 out-of-sample evaluation is not yet reported; therefore, this version makes no claim of forecast superiority.

Keywords: financial forecasting; nonstationary time series; Mamba; state-space models; multi-scale analysis

1 Introduction

Financial level forecasting is difficult not simply because prices and rates are persistent, but because the representation useful in one local window can become poorly scaled or emphasize the

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wrong time scale in another. Location, volatility, and cross-series dependence can all change while the prediction target remains a level. An architecture for this setting should therefore preserve local scale information, separate slow and rapidly changing structure without treating either as an identified economic mechanism, and retain temporal order at more than one resolution.

Mamba supplies an input-selective state-space recurrence with a linear recurrent scan [1]. However, the tokenization choice determines what that scan can model. In an inverted variable-token encoder, each whole variable history is embedded into one token [2, 3]. For a univariate task, the resulting token sequence has length one: the Mamba block can transform the embedded history, but it cannot perform a recurrence across temporal tokens. This implementation-level limitation motivates a temporal-patch state-space model rather than another variable-token variant.

We construct PMS-Mamba (patch multi-scale Mamba), which applies reversible window normalization, an additive moving-average decomposition, mean-pooled temporal resolutions, and bidirectional Mamba encoders over ordered patches. The model is designed around three representation hypotheses: slow and high-frequency variation should not be forced through the same branch; a local perturbation should be smoothed rather than amplified before coarse-scale encoding; and a univariate state-space model should receive a sequence of temporal patches rather than a single variable token. These are architectural hypotheses, not assumptions that financial markets follow a fixed trend–seasonal–residual process.

The four financial panels are used first to test whether the proposed representation problem is present in the data. They span foreign exchange, Treasury yields, cross-asset prices, and cryptoassets, and produce paired target-only and multivariate-input tasks. The analysis uses only training and validation observations to characterize nonstationarity and distribution shift; it does not use the protected 2023–2026 targets to select the narrative.

The contributions are:

- an implementation-grounded diagnosis of the single-token limitation of inverted Mamba encoders for univariate forecasting;
- PMS-Mamba, which joins additive decomposition, multi-scale pooling, temporal patches, and bidirectional selective state-space encoding in one financial-level forecaster;
- representation-level propositions for additive component preservation, pooling stability, and bounded fixed scale fusion, with proofs in Appendix A; and
- a data analysis across four financial panels together with a leakage-controlled evaluation design for testing the model against persistence, statistical, and neural references.

This version reports the model, its representation properties, and selection-safe data analysis. The final out-of-sample outcome matrix is not yet available, so the manuscript makes no claim that PMS-Mamba improves forecasting accuracy.

2 Related Work

2.1 Nonstationary Representations for Forecasting

Forecasting under distribution shift has motivated both reversible normalization and architecture-level responses. RevIN normalizes each input instance and reverses the transformation at the output [4]; Non-stationary Transformers add a learned de-stationary mechanism because normalization can also remove predictive information [5]. Decomposition methods separate slow and residual structure so that a single representation need not explain every temporal scale [6, 7, 8]. PatchTST, N-BEATS,

and N-HiTS provide complementary examples of patching, basis expansion, and explicit multi-rate temporal structure [9, 10, 11]. PMS-Mamba uses these ideas as representation choices and does not interpret its components as uniquely identified financial processes.

2.2 Mamba Tokenization for Time Series

Mamba replaces self-attention with an input-selective state-space recurrence [1]. S-Mamba applies bidirectional Mamba blocks over variable tokens [2]; Bi-Mamba+, MambaTS, and TimeMachine study bidirectionality, scan directions, and multiple Mamba pathways [12, 13, 14]. The distinction between variable tokens and temporal tokens is central in the present work. Inverted representations can capture cross-variable relations when several variables are available [3], but they reduce a target-only input to one token. PMS-Mamba instead makes its selective state-space encoder operate on temporal patch sequences at each scale.

Several recent studies also combine Mamba with multiple scales or decomposition. ms-Mamba uses several temporal sampling rates [15]; KARMA combines multilevel decomposition with multi-scale Mamba blocks [16]; and DMamba assigns different modules to decomposed components [17]. The contribution here is not a priority claim for these individual ideas. It is a specific patch-based composition, a direct account of why it is needed for univariate Mamba inputs, and an evaluation focused on nonstationary financial levels.

2.3 Financial Data Analysis and Evaluation

Financial time-series studies are sensitive to the definition of the target, temporal split, regime coverage, and reference forecast [18]. For persistent levels, last-value, drift, and classical univariate or multivariate forecasts are necessary references; forecasting competitions also show that simple methods can remain competitive across horizons [19, 20]. The empirical design therefore reports results by target and horizon rather than pooling raw errors with different units. Repeated neural fits and paired loss comparisons over common forecast origins quantify uncertainty, but they do not establish a stable market mechanism, economic value, or trading profitability.

3 PMS-Mamba

3.1 Forecasting Task and Notation

Let $\mathbf{X}_{t-L+1:t} \in \mathbb{R}^{L\times C}$ be the observations available at forecast origin t, and let $\mathbf{Y}_{t+1:t+H} \in \mathbb{R}^{H}$ be the future path of one target series. The target-only task uses $C = 1$. The paired multivariate-input task uses all $C > 1$ panel variables while evaluating only the target stored in the final channel. Thus, the multivariate task tests whether contemporaneously available histories improve one target forecast; it is not a claim of unrestricted multivariate forecasting or causal cross-market influence. Separate models are fitted for horizons

$$H \in \{1, 5, 20, 60, 120, 252\}. \tag{1}$$

3.2 Why Temporal Patches Are Needed

Mamba is an input-selective state-space model with updates

$$\mathbf{h}_j = \bar{\mathbf{A}}_j \mathbf{h}_{j-1} + \bar{\mathbf{B}}_j \mathbf{e}_j, \qquad \mathbf{o}_j = \mathbf{C}_j \mathbf{h}_j, \tag{2}$$

where the transition and readout parameters depend on the input token $\mathbf{e}_j$ [1]. The recurrence is only meaningful along the sequence presented to it. S-Mamba uses an inverted embedding that maps each complete variable history to one token [2]. It therefore receives C tokens rather than L temporal tokens.

Proposition 1 (Single-token limitation)**.** *For a target-only input ($C = 1$), an inverted variable-token encoder presents exactly one token to its state-space block. Its output may depend on the complete input history through the embedding, but the state-space block has no transition between distinct temporal tokens.*

Proposition 1 does not imply that S-Mamba is incapable of forecasting a univariate series; its embedding and projection remain learned functions of the input window. It identifies a mismatch between a temporal state-space motivation and a one-token implementation. PMS-Mamba resolves that mismatch by tokenizing each normalized time history into ordered temporal patches.

3.3 Decomposition-Aware Multi-Scale Patch Encoder

For every input window, PMS-Mamba first applies reversible per-window normalization. For a batch item b and channel c,

$$\mu_{b,c} = \frac{1}{L}\sum_{t=1}^{L} X_{b,t,c}, \qquad \sigma_{b,c} = \sqrt{\frac{1}{L}\sum_{t=1}^{L}(X_{b,t,c} - \mu_{b,c})^2 + \epsilon}, \tag{3}$$

$$Z_{b,t,c} = \frac{X_{b,t,c} - \mu_{b,c}}{\sigma_{b,c}}. \tag{4}$$

The final forecast is returned to the input window's location and scale. This operation stabilizes the representation under local changes in level and magnitude; because $\epsilon > 0$, it is not claimed to provide exact affine equivariance.

Let $\mathcal{M}_m$ denote the boundary-replicated moving average with window m used in the implementation. PMS-Mamba constructs three additive components,

$$\mathbf{T} = \mathcal{M}_m(\mathbf{Z}), \tag{5}$$

$$\mathbf{D} = \mathbf{Z} - \mathbf{T}, \tag{6}$$

$$\mathbf{S} = \mathcal{M}_m(\mathbf{D}), \tag{7}$$

$$\mathbf{R} = \mathbf{D} - \mathbf{S}. \tag{8}$$

The symbols $\mathbf{T}$, $\mathbf{S}$, and $\mathbf{R}$ denote a smoothed component, a smoothed residual, and a residual component, respectively. They are architectural representations, not identified trend, seasonal, or structural financial processes.

For each scale $s \in \mathcal{S} = \{1, 2, 4\}$, average pooling constructs

$$\mathbf{V}^{(s)} = \mathcal{P}_s(\mathrm{concat}_C[\mathbf{T}, \mathbf{S}, \mathbf{R}])\,. \tag{9}$$

Each channel of $\mathbf{V}^{(s)}$ is divided into ordered, right-padded patches of length p and stride r. The patch embedding is followed by a bidirectional Mamba encoder and a flattened projection head. If g_s denotes this scale-specific mapping, its output is regrouped into the three components and summed within each original variable:

$$\widetilde{\mathbf{Y}}_c^{(s)} = \sum_{k\in\{T,S,R\}} g_{s,k,c}(\mathbf{V}^{(s)}), \qquad \widetilde{\mathbf{Y}} = \frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}} \widetilde{\mathbf{Y}}^{(s)}. \tag{10}$$

The frozen PMS-Mamba-PD configuration uses this fixed mean scale fusion rather than learned or shift-gated fusion. Figure 1 summarizes the implemented flow.

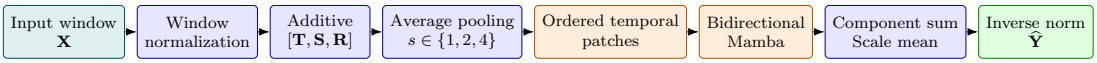


Figure 1: PMS-Mamba-PD. A normalized input window is decomposed, pooled at multiple temporal resolutions, converted into ordered patches, and processed by a bidirectional Mamba encoder at each scale. Component forecasts are summed and scale forecasts are mean fused before inverse normalization.

3.4 Representation Properties

The following propositions concern the fixed representation and aggregation operations in PMS-Mamba, not the unknown data-generating process or the nonlinear learned branches.

Proposition 2 (Additive component preservation)**.** *For every normalized input window* $\mathbf{Z}$*, the implemented decomposition satisfies*

$$\mathbf{Z} = \mathbf{T} + \mathbf{S} + \mathbf{R}. \tag{11}$$

Proposition 3 (Stable pooling and bounded mean fusion)**.** *For any two component tensors* $\mathbf{U}$ *and* $\mathbf{V}$*, every average-pooling operator used by PMS-Mamba satisfies*

$$\|\mathcal{P}_s(\mathbf{U}) - \mathcal{P}_s(\mathbf{V})\|_\infty \leq \|\mathbf{U} - \mathbf{V}\|_\infty . \tag{12}$$

For fixed scale-specific forecasts, each coordinate of fixed mean fusion lies between the minimum and maximum of that coordinate over the scales.

Proofs appear in Appendix A. The first property prevents information loss before learned component branches; the second shows that pooling and fixed fusion do not amplify a coordinatewise perturbation by themselves. Neither property establishes forecast accuracy, and both require empirical evaluation of the learned maps g_s.

3.5 Ablation Logic and Comparators

The implemented Mamba configurations isolate progressively richer representations: S-Mamba uses one inverted variable-token path; NS-Mamba-D adds the additive decomposition; MS-Mamba-MD adds multi-scale component tokens; and PMS-Mamba-PD replaces the variable-token encoder with temporal patch encoders. These comparisons do not constitute a complete factorial ablation because decomposition, scale construction, and tokenization change jointly in some transitions. The recomputed study will therefore add matched patch-only, decomposition-only, multi-scale-only, and full PMS-Mamba variants with parameter counts reported for every model.

The external learned comparators are DLinear, NLinear, PatchTST, TimesNet, and iTransformer [8, 9, 21, 3]. Last-value, random-walk-with-drift, and ARIMA provide univariate references; VAR on first differences provides a multivariate reference. ARIMA orders (p, d, q) are selected on training data by AIC over $d \in \{0, 1\}$ and $p, q \in \{0, 1, 2, 3\}$, excluding ARIMA$(0, 0, 0)$. For multivariate tasks, VAR lags from 1 through 10 are selected by training BIC and forecasts are accumulated from differences to levels. The statistical reference coefficients are estimated once from all pre-test training and validation observations and their states are updated sequentially with observed values; neural weights use training observations while validation controls early stopping.

Table 1: Frozen financial panels. Counts refer to target observations within each calendar split. Each panel defines a target-only S task and a complete-panel MS task.

Panel	Date range	Inputs in MS mode (target last)	Target	Total	Train	Validation	Test
Foreign exchange	1999-01-04–2026-06-30	EURUSD, CADUSD, JPYUSD, CHFUSD, AUDUSD, GBPUSD	GBPUSD	6,894	5,521	499	874
Treasury yields	2006-02-09–2026-06-30	DGS2, DGS5, DGS30, DGS10	DGS10	5,101	3,728	500	873
Cross asset	2016-07-25–2026-06-30	S&P 500, WTI, VIX, GLD	GLD	2,483	1,112	502	869
Cryptoasset	2018-05-04–2026-06-30	ETHUSDT, BNBUSDT, XRPUSDT, BTCUSDT	BTCUSDT	2,980	973	730	1,277

3.6 Training Objective

All learned models minimize target-channel mean squared error on the scaled representation,

$$\mathcal{L} = \frac{1}{BH} \sum_{b=1}^{B} \sum_{h=1}^{H} \left(\hat{Y}_{b,h} - Y_{b,h} \right)^2 . \tag{13}$$

Forecasts and targets are transformed back to the target's original scale before evaluation.

4 Financial Data Analysis and Evaluation Design

4.1 Financial Panels and Data Provenance

The frozen data cutoff is 2026-06-30. Table 1 describes four daily panels after chronological sorting and complete-case intersection. The foreign-exchange panel uses six FRED exchange-rate series, with reciprocal transformations where needed so that every column is quoted as U.S. dollars per unit of foreign currency [22]. The yield panel contains 2-, 5-, 10-, and 30-year U.S. Treasury constant-maturity rates from FRED, with DGS10 as target [22]. The cross-asset panel aligns FRED S&P 500, WTI crude-oil, and VIX levels with the unadjusted GLD close obtained from Yahoo Finance [22, 23]. The cryptoasset panel uses daily UTC spot-closing prices from the official Binance public archive [24]. Because redistribution requirements differ across source providers, data access and any public release remain subject to the corresponding source terms.

All processed panels have unique increasing dates and no missing cells. Complete-case alignment avoids imputed targets but produces an irregular observation calendar governed by common market availability; horizons are therefore measured in panel observations, not elapsed calendar days. The fixed splits are training through 2020-12-31, validation from 2021-01-01 through 2022-12-31, and final evaluation from 2023-01-01 through 2026-06-30. Validation and test loaders can use preceding observations as input context, but every forecast target remains inside its named split. Each scaler is fitted only on the training rows.

The legacy GLD and Exchange comparison settings were used during model development and are excluded from the present evidence boundary. Non-financial application datasets are also outside the protocol because the research question is specifically nonstationary financial forecasting rather than broad application-domain ranking. These exclusions were frozen with the new protocol and are not based on outcomes from its protected final period.

4.2 Selection-Safe Data Analysis of Nonstationarity and Shift

Before test access, diagnostics read only training and validation targets. For training levels, the augmented Dickey–Fuller (ADF) test includes a constant and trend, and the KPSS test uses trend stationarity as its null [25, 26]. The same tests are applied to first differences with a constant but no trend. The diagnostic pair is interpreted descriptively: failure to reject an ADF unit-root null

Table 2: Selection-safe diagnostics computed without reading final-period targets. Mean shift is measured in training standard deviations; standard-deviation and difference-volatility columns are validation-to-training ratios. Correlation shift is the largest absolute change in a feature–target first-difference correlation.

Panel	Level ADF p	Level KPSS p	Mean shift	SD ratio	Difference-volatility ratio	Max. correlation shift
Foreign exchange	0.682	≤ 0.010	-1.294	0.430	0.823	0.268
Treasury yields	0.363	≤ 0.010	-0.512	0.858	1.129	0.026
Cross asset	0.719	≤ 0.010	1.708	0.344	1.251	0.323
Cryptoasset	1.000	≤ 0.010	7.750	3.669	4.521	0.106

together with rejection of the KPSS trend-stationarity null supports treating the observed level as nonstationary, but neither test identifies a unique data-generating mechanism.

Table 2 reports the prespecified level tests and validation shifts. All four training targets show ADF $p > 0.36$ and KPSS $p \leq 0.01$ for levels. The validation period also changes materially in at least one location, scale, difference-volatility, or cross-series-correlation summary for every panel. Cryptoassets show the largest location and difference-volatility shifts, while cross-asset and foreign-exchange panels show the largest maximum absolute changes in feature–target first-difference correlations. These patterns do not identify an economic mechanism or imply that a multi-scale model will be more accurate. They establish a concrete representation problem for the model: local level and scale are not stable across the observed periods, high-frequency variation changes unevenly by panel, and cross-series information is not invariant from training to validation. The target-only PMS-Mamba design addresses the first two properties without requiring cross-series inputs; the paired multivariate task tests the third property empirically.

Figure 2 visualizes the same evidence boundary. Targets are standardized by training means and standard deviations; trailing 60-observation first-difference volatility is divided by its training median. The gray interval is validation, and no 2023–2026 target is shown.

4.3 Model Selection and Held-Out Evaluation

The learned set contains DLinear, NLinear, PatchTST, TimesNet, iTransformer, S-Mamba, NS-Mamba-D, MS-Mamba-MD, and PMS-Mamba-PD. Every model–task study receives 12 effective Optuna trials, where completed and pruned trials count toward the fixed budget. A seeded TPE sampler evaluates calibration horizons 5, 60, and 252; median pruning begins after four startup trials. The three best completed calibration trials are then replayed across all six validation horizons. The configuration with the smallest balanced validation RMSSE is frozen for that model and task. Training uses Adam, at most 10 epochs, early-stopping patience 3, and mean squared error. Model-specific search spaces are listed in Appendix B.1.

No final-period evaluation is permitted during selection. The selection audit verifies task and model coverage, protocol and data hashes, horizon lists, validation flags, and that the final-evaluation flag remains unopened for every selected record. Only after this audit may the explicit `--unlock-test` gate be invoked. The selected neural configurations are then retrained and evaluated with seeds 2023–2027. Neural weights are fitted on training observations, with validation observations used for early stopping rather than gradient updates. The complete neural matrix contains $8 \times 9 \times 5 = 360$ task–model–seed cells, each comprising six separately fitted horizon models. Statistical references refuse to access the final period until all 360 neural cells are complete.

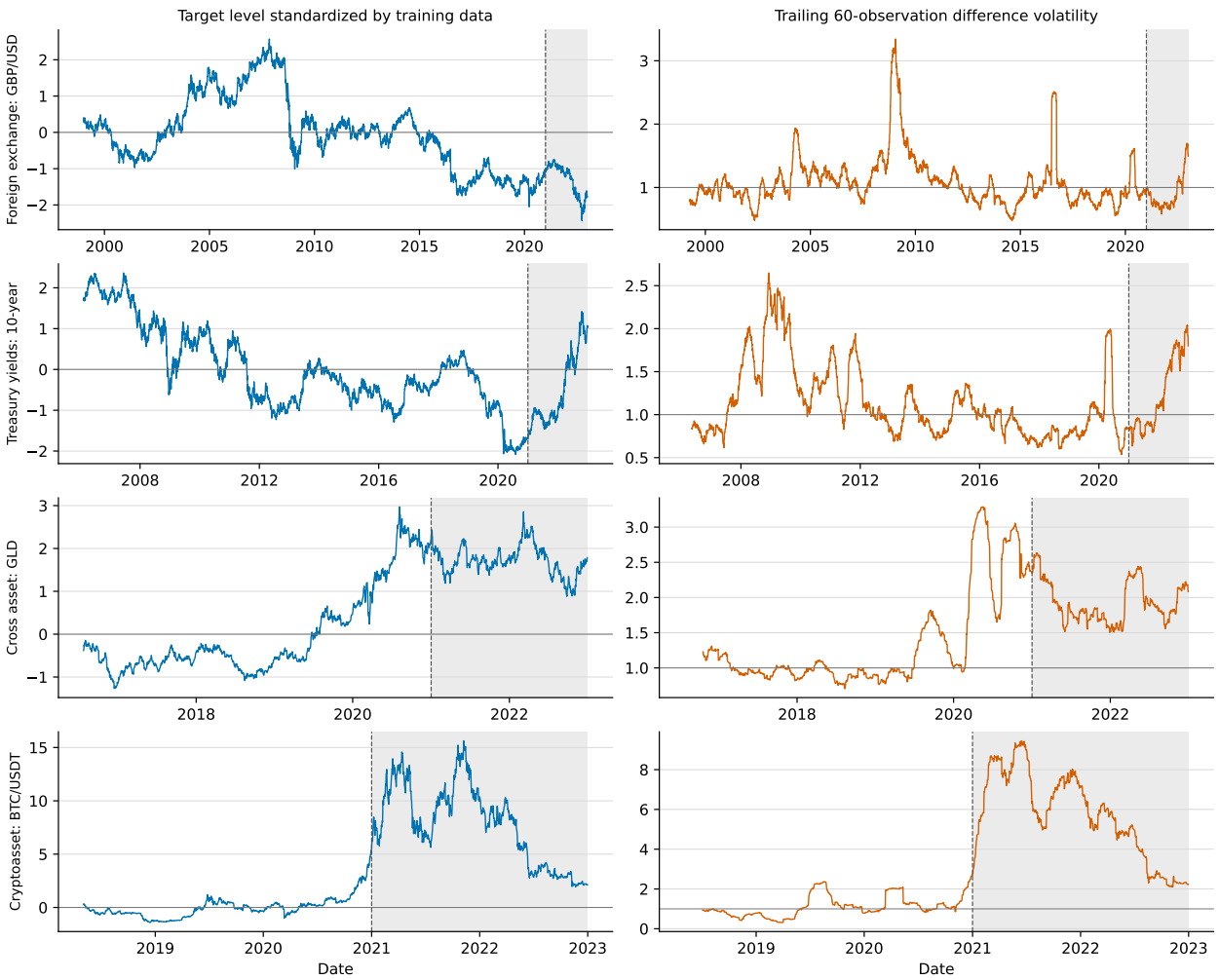


Figure 2: Selection-safe financial data characterization. Dashed vertical lines mark the start of validation, and gray regions denote validation observations. The left column shows target levels standardized using training data only. The right column shows lagged trailing 60-observation first-difference volatility relative to its training median. Test targets are neither read nor plotted.

4.4 Metrics and Prespecified Analysis

For task d, horizon H, forecast origin o, and lead h, let $e_{d,o,h} = Y_{d,o,h} - \hat{Y}_{d,o,h}$. Path MSE averages $e^2_{d,o,h}$ over every available origin and all leads $1, \ldots, H$; terminal MSE uses only lead H. MAE and RMSE are defined analogously. RMSSE divides MSE by the mean squared one-step difference of the inverse-transformed training target and then takes the square root [20]:

$$\mathrm{RMSSE}_{d,H} = \sqrt{\frac{\mathrm{MSE}_{d,H}}{(T_d - 1)^{-1} \sum_{t=2}^{T_d} (Y_{d,t} - Y_{d,t-1})^2}}. \tag{14}$$

Metrics remain task-specific; raw errors are never averaged across targets with different units. Balanced RMSSE is the unweighted mean over the six horizons, equivalently the mean of the short $(1, 5)$, middle $(20, 60)$, and long $(120, 252)$ group means.

For each fitted model, skill relative to last-value is

$$\mathrm{Skill}_{d,H} = 1 - \frac{\mathrm{MSE}_{d,H}^{\mathrm{model}}}{\mathrm{MSE}_{d,H}^{\mathrm{LV}}}. \tag{15}$$

Positive values indicate lower path MSE than persistence. Neural metrics are summarized as mean and standard deviation over five seeds. S and MS estimates are paired by panel, model, and horizon.

For confirmatory contrasts, origin-level squared losses are first averaged across the five neural seeds. Each proposed variant (NS-Mamba-D, MS-Mamba-MD, and PMS-Mamba-PD) is paired with S-Mamba and with the non-Mamba model selected on validation for the same task. A circular moving-block bootstrap resamples the paired origin-level loss difference 10,000 times. The block length is $\min\{n, \max(H, \lceil n^{1/3} \rceil)\}$, and percentile 95% intervals are reported. Within each task, proposal, and comparator, two-sided bootstrap p-values across the six horizons receive Holm adjustment.

The prespecified secondary regime analysis assigns each origin using the standard deviation of target first differences over the preceding 60 observations, lagged by one observation. Training-period 0.25 and 0.75 quantiles define low-, middle-, and high-volatility regimes. Return forecasting is deferred until the primary level study is complete and cannot replace an unfavorable level result.

4.5 Reproducibility and Evidence Boundary

The analysis plan and data cutoff were recorded before final-outcome analysis on 2026-07-25. The study record retains the exact preprocessing, chronological splits, model configurations, software settings, and execution logs needed to reproduce the completed analyses.

At the time of this manuscript version, validation-only selection has completed and passed its coverage audit, but the final outcome files do not exist. Accordingly, no validation ranking is presented as a result, no test target has been opened for manuscript analysis, and all performance statements remain pending.

5 Financial Data Analysis and Current Empirical Status

5.1 Observed Representation Shift

The available evidence describes the modeling problem targeted by PMS-Mamba but does not yield a conclusion about forecast accuracy. Across the four training targets, the level ADF test fails to reject a unit root while the level KPSS test rejects trend stationarity at its reported lower p-value bound (Table 2). This pattern does not prove that the series share one integrated data-generating process; it shows that a common stationary-level representation is not adequate for these observed histories.

The validation period differs materially from training (Figure 2). Foreign exchange and Treasury yields shift downward by 1.294 and 0.512 training standard deviations, whereas GLD shifts upward by 1.708 training standard deviations. BTCUSDT exhibits the largest changes: its validation mean is 7.750 training standard deviations above the training mean, its level standard deviation is 3.669 times the training value, and its first-difference standard deviation is 4.521 times the training value. The range of volatility changes is therefore not reducible to one global rescaling rule.

Cross-series structure also changes between training and validation. The largest absolute change in a feature–target first-difference correlation is 0.268 for foreign exchange, 0.026 for yields, 0.323 for cross assets, and 0.106 for cryptoassets. This variation supports evaluating target-only and

multivariate inputs separately: additional variables may be informative in one panel or period but cannot be assumed to help uniformly.

5.2 Mechanism-Oriented Evaluation Status

The current study has completed source retrieval, preprocessing, checksums, selection-safe data analysis, and validation-only hyperparameter selection. The selection audit passed for every task and model. The protected 2023–2026 neural matrix, statistical-reference forecasts, paired loss contrasts, and final-period diagnostics have not yet been generated. Table 3 records this boundary.

Table 3: Evidence status for the current manuscript. "Locked" means that the workflow requires an explicit gate and a preceding completeness audit before reading final-period targets.

Evidence component	Status	Admissible use in this version
Source and processed-data manifests	Complete	Data provenance and descriptive analysis
Training/validation diagnostics	Complete	Representation motivation
Validation-only model selection	Complete (audit passed)	Configuration freeze only; no ranking claim
2023–2026 neural evaluation	Locked	None
Statistical-reference evaluation	Not generated	None
Bootstrap and regime outcome analysis	Not generated	None

The correct empirical question is not whether one Mamba label has the smallest isolated point estimate. It is whether temporal patching, decomposition, and multi-scale pooling each contribute beyond matched alternatives, and whether any contribution persists against last-value and statistical references across panels, horizons, and volatility regimes. No such conclusion is possible from the current evidence.

5.3 Recomputation Plan for the Model Claims

The next calculation cycle will preserve the chronological splits and final-period gate, but will replace the current partial comparison with a mechanism-oriented ablation matrix. It will include a target-only variable-token base, a patch-only model, a decomposition-only model, a multi-scale-only model, decomposition-plus-multi-scale variants, and full PMS-Mamba with matched parameter counts and the same selection budget. For each task and horizon, the final report will include path and terminal RMSSE, last-value skill, five-seed uncertainty, origin-level moving-block bootstrap intervals, and panel-specific failures as well as improvements.

The paired S/MS analysis will remain separate from the architectural ablation: it tests the incremental predictive value of the supplied panel histories, not the mechanism of PMS-Mamba. Return forecasting, if later added, will be labelled as a robustness analysis and cannot replace a level-forecast result.

6 Discussion and Limitations

The model is motivated by an implementation fact and by observed data variation. In a target-only inverted Mamba encoder, the state-space block receives one variable token; PMS-Mamba instead applies its state-space scan to ordered temporal patches. The financial data analysis then shows

that location, scale, difference volatility, and cross-series correlation can change materially between training and validation. Together these observations support studying a decomposition-aware multi-scale temporal representation, but they do not establish that it is optimal or economically structural.

The theory is deliberately limited. Additive component preservation ensures that the fixed decomposition itself does not discard the normalized input, and non-expansive pooling plus mean fusion constrain only the preprocessing and aggregation stages. The learned patch embeddings, Mamba branches, and projection heads are nonlinear and their forecast error, stability, and calibration require empirical measurement. The components should therefore not be interpreted as an identified trend, seasonal process, or market factor.

The redesigned empirical analysis must distinguish model mechanisms. The present four Mamba configurations do not isolate every change in tokenization, decomposition, scale construction, and parameter count. A new calculation cycle is required to run a factorial ablation with matched capacities, fixed data splits, and equal selection budgets. The central result should report when the patch representation helps, when decomposition or pooling helps, and when none helps, rather than collapse all tasks into a single rank.

The data scope also remains limited. Four daily panels cannot represent every market, sampling frequency, or institutional regime. Complete-case alignment avoids imputing targets but creates an irregular observation calendar governed by common market availability. The shortest cryptoasset training segment contains 973 observations, and one final period cannot establish temporal generalization across all historical regimes. Rolling-origin sensitivity analysis is therefore desirable after the primary held-out calculation.

Finally, point-forecast accuracy is not evidence of profitable trading, market inefficiency, directional accuracy, calibrated risk, or robustness to transaction costs. Final claims must be anchored to last-value, drift, ARIMA, and VAR-on-differences references; use task-specific metrics; account for overlapping forecast origins; and preserve null or unfavorable results. Dataset redistribution terms, a citable code archive, author contributions, funding, competing interests, and venue-specific AI-use disclosure still require accountable author confirmation before submission.

7 Conclusion

This work develops PMS-Mamba, a decomposition-aware multi-scale patch Mamba model for non-stationary financial level forecasting. Its central design choice is to place the selective state-space scan over ordered temporal patches rather than over variable tokens alone. The resulting representation preserves the normalized input through additive decomposition, pools temporal variation without amplifying a maximum-norm perturbation, and uses bounded fixed mean fusion across scales.

Selection-safe analysis of foreign exchange, Treasury yields, cross-asset prices, and cryptoassets documents material changes in local level, scale, volatility, and cross-series dependence. These observations motivate the model but do not demonstrate forecast superiority. The required next step is a recomputed, mechanism-oriented ablation study with matched capacity and a protected out-of-sample evaluation. Until those results are complete, the manuscript makes no claim that PMS-Mamba improves financial forecasts.

A Theoretical Properties

A.1 Proof of Proposition 1

The inverted embedding constructs one token for each of the C input variables and passes the resulting sequence to the bidirectional Mamba encoder. When $C = 1$, the encoder sequence is $(\mathbf{e}_1)$. There is no pair of distinct temporal token indices $j-1$ and j at which a transition can be evaluated. The output can still depend on the complete history because $\mathbf{e}_1$ is computed from that history, but the state-space block has no recurrence across temporal tokens. □

A.2 Proof of Proposition 2

By construction, $\mathbf{D} = \mathbf{Z} - \mathbf{T}$ and $\mathbf{R} = \mathbf{D} - \mathbf{S}$. Therefore,

$$\mathbf{T} + \mathbf{S} + \mathbf{R} = \mathbf{T} + \mathbf{S} + (\mathbf{Z} - \mathbf{T}) - \mathbf{S} = \mathbf{Z}. \tag{16}$$

The result holds independently of the moving-average width and boundary replication rule. □

A.3 Proof of Proposition 3

Every coordinate of $\mathcal{P}_s(\mathbf{U})$ is an arithmetic mean of a nonempty set of input coordinates. For any pooled window I,

$$\left| \frac{1}{|I|} \sum_{i \in I} (U_i - V_i) \right| \leq \frac{1}{|I|} \sum_{i \in I} |U_i - V_i| \leq \|\mathbf{U} - \mathbf{V}\|_\infty. \tag{17}$$

Taking the maximum over pooled coordinates yields the pooling inequality. For fixed forecasts $\widetilde{Y}^{(s)}$, mean fusion is a convex combination with weights $1/|\mathcal{S}|$. Each fused coordinate therefore lies in the interval spanned by the corresponding scale-specific coordinates. □

B Frozen Protocol and Reproducibility Details

B.1 Frozen protocol

Table 4: Controlling protocol fields.

Field	Frozen value
Version	`financial_confirmatory_v1`
Frozen on	2026-07-25
Data cutoff	2026-06-30
Train / validation / test	through 2020 / 2021–2022 / 2023–2026-06
Horizons	1, 5, 20, 60, 120, 252 observations
Calibration horizons	5, 60, 252 observations
Trials / full replays	12 effective trials / top 3 completed trials
Training cap / patience	10 epochs / 3 epochs
HPO sampler seed	20260725
Final neural seeds	2023, 2024, 2025, 2026, 2027
Bootstrap	10,000 circular moving-block replicates; seed 20260725

The eight task identifiers are listed by panel in Table 5.

Table 6: Non-Mamba categorical search spaces. "LR" denotes learning rate and "BS" batch size.

Model	Lookback L	Width choices	LR	BS	Specific choices
DLinear	96, 192, 256	–	0.003, 0.005	32, 64	Moving average 13, 25; shared channels
NLinear	64, 192, 320, 384	–	0.005, 0.01	32, 64	Shared channels
PatchTST	128, 256, 320	d_m: 128, 256; d_{ff}: 512, 2048	0.0002, 0.0003, 0.0005	16, 32	Patch/stride: 16/8, 24/12, 32/16
TimesNet	128, 256, 320	d_m: 32, 64; d_{ff}: 128, 256	0.0001, 0.0002, 0.0003	16, 32	Top-$k = 5$; kernels 6, 8
iTransformer	128, 192, 256	d_m: 128, 256; d_{ff}: 256, 512	0.0001, 0.0002, 0.0003	16	Reversible normalization on

Table 5: Paired task identifiers.

Panel	S task	MS task
Foreign exchange	`financial_fx_s`	`financial_fx_ms`
Treasury yields	`financial_yields_s`	`financial_yields_ms`
Cross asset	`financial_cross_asset_s`	`financial_cross_asset_ms`
Cryptoasset	`financial_crypto_s`	`financial_crypto_ms`

Every manifest entry stores its processed-data hash, the protocol hash, fixed split dates, target, input/output channel counts, horizon suite, and term groups.

B.2 Neural search spaces

The following values are categorical choices, not continuous ranges. All models use two encoder layers, dropout 0.1, and label length $L/2$ unless the implementation ignores a field. The common Mamba space is $L \in \{16, 32, 64, 96, 128, 192, 256, 320, 384\}$, $d_{\text{model}} \in \{64, 128\}$, $d_{\text{ff}} \in \{128, 256\}$, state dimension $\in \{16, 32\}$, moving-average window $\in \{13, 25, 49\}$, learning rate $\in \{5 \times 10^{-5}, 10^{-4}, 2 \times 10^{-4}, 3 \times 10^{-4}\}$, and batch size 16. S-Mamba adds no variant-specific choice. NS-Mamba-D fixes standard rather than adaptive normalization, activates decomposition, and disables shift gating. MS-Mamba-MD chooses scales from $\{1, 2, 4\}$, $\{1, 2, 4, 8\}$, $\{1, 3, 6\}$, or $\{1, 4, 8\}$; it fixes decomposition on, mean scale fusion, and no linear skip. PMS-Mamba-PD chooses $\{1, 2, 4\}$ or $\{1, 2, 4, 8\}$ and a patch-length/stride pair from $(8, 4)$, $(8, 8)$, $(16, 4)$, or $(16, 8)$; infeasible scaled patch layouts are rejected before training.

Optuna uses TPE with the frozen sampler seed and a median pruner with four startup trials, one warm-up step, and one-step intervals. Interrupted `RUNNING` trials are marked `FAIL` on recovery. The fixed budget counts `COMPLETE` and `PRUNED` trials; failed trials do not consume it. Only completed trials can enter the top-three full validation replay.

B.3 Reference-model specification

Last-value and random-walk-with-drift have no fitted hyperparameters. ARIMA searches $d \in \{0, 1\}$ and $p, q \in \{0, 1, 2, 3\}$ on the training target and selects minimum AIC; ARIMA$(0, 0, 0)$ is excluded. A time trend is used for $d = 1$, and a constant plus trend is used for $d = 0$. VAR-difference is available only for MS tasks and selects lags 1–10 by minimum training BIC with constant and trend. After order or lag selection, coefficients are estimated once using all observations before the test boundary. At each final origin, ARIMA state and VAR lagged differences use observations available up to that origin, while fitted coefficients remain fixed.

B.4 Artifact Gates and Regeneration

The repository workflow runs hyperparameter selection, selection audit, protected final evaluation, statistical-reference evaluation, and result analysis as separate stages in the Anaconda `mamba` environment. The exact commands, configuration hashes, and machine-readable outputs are retained with the code release rather than embedded in the manuscript. The selection command refuses missing models, wrong hashes or horizons, non-validation selection, and prior test access. The final runner records task, model, seed, horizon, origin dates, predictions, targets, run time, peak GPU memory, software versions, GPU identity, and deterministic-algorithm settings. Completed cells are written atomically and skipped on restart.

The statistical runner is blocked until every neural cell is complete. The analysis script then verifies forecast shapes, origin dates, saved metrics, metadata, and expected row counts before writing aggregate metrics, S/MS pairs, regime metrics, and bootstrap contrasts. The manuscript generator additionally requires full-period diagnostics marked `test_targets_read=true` and exactly 10,000 bootstrap replicates. No partial task selection can produce the final result section.

The current draft instead uses only the selection-safe diagnostics manifest. That manifest records a training/validation scope and confirms that test targets were not read. The final generated tables and copied machine-readable supplements will be added only after the protected workflow completes.

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